and vaccines. In media and entertainment, drones have redefined cinematography, allowing for aerial perspectives previously achievable only through costly manned aircraft. Furthermore, the increasing adoption of UAVs in urban planning, traffic analysis, and law enforcement demonstrates their growing integration into smart city frameworks.

Figure 1.2 illustrates the expanding role of drones across commercial and civilian sectors, emphasizing how unmanned aerial vehicles (UAVs) have become essential tools in industries that demand precision, efficiency, and accessibility. At the center of the figure is a multi-rotor drone, represent-ing the technological versatility that supports diverse applications, including aerial photography, infrastructure inspection, agriculture, and environmental monitoring.